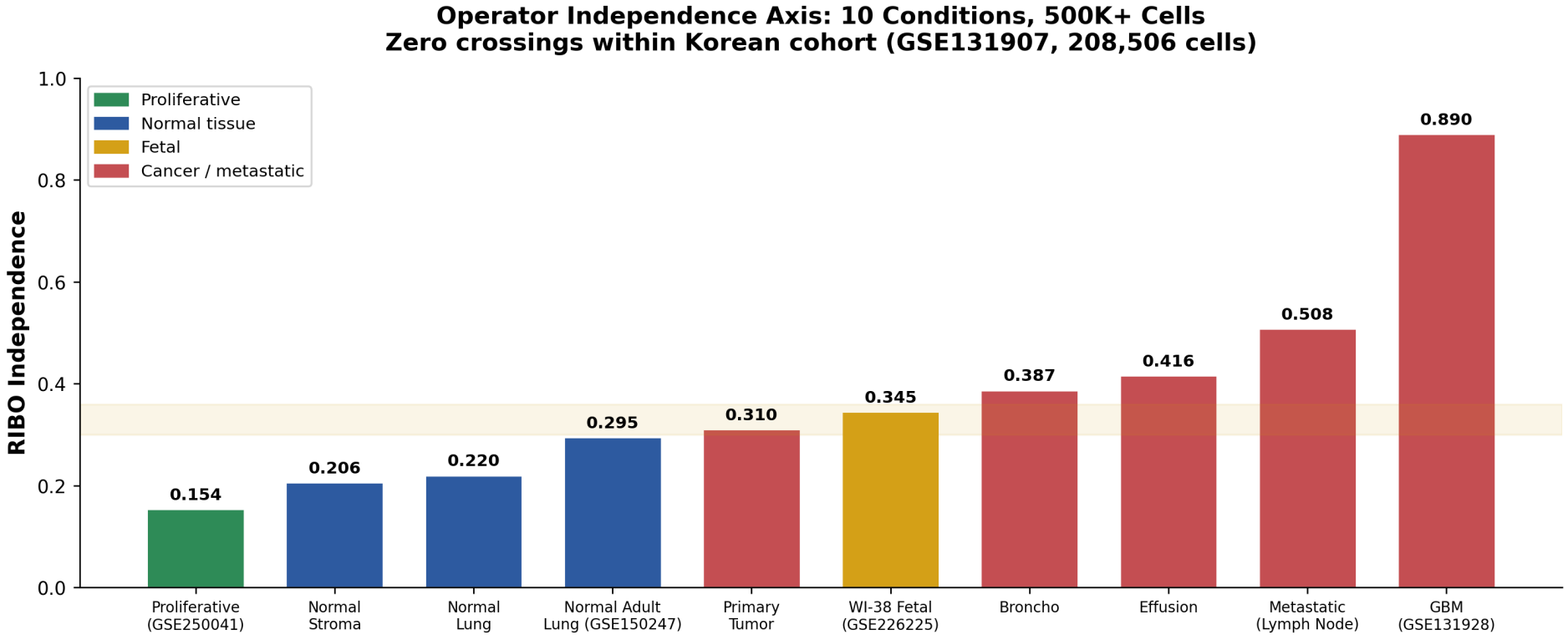
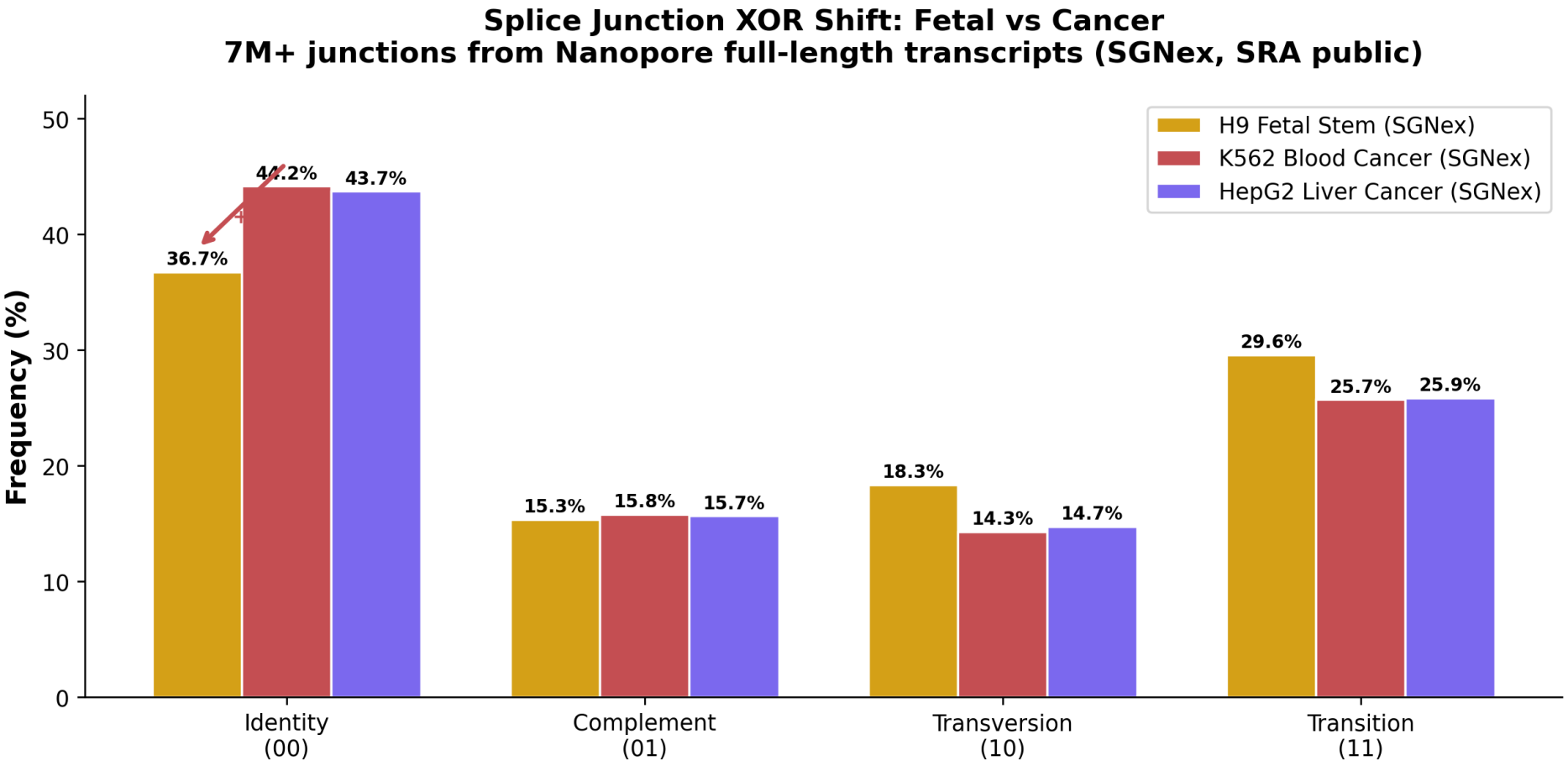
Figure 1. Operator Independence Axis



10 conditions, 500K+ cells. Seven Korean NSCLC conditions (GSE131907, 208,506 cells) show zero crossings. Proliferative (0.154) through normal tissue through primary tumor (0.310) to GBM (0.890).

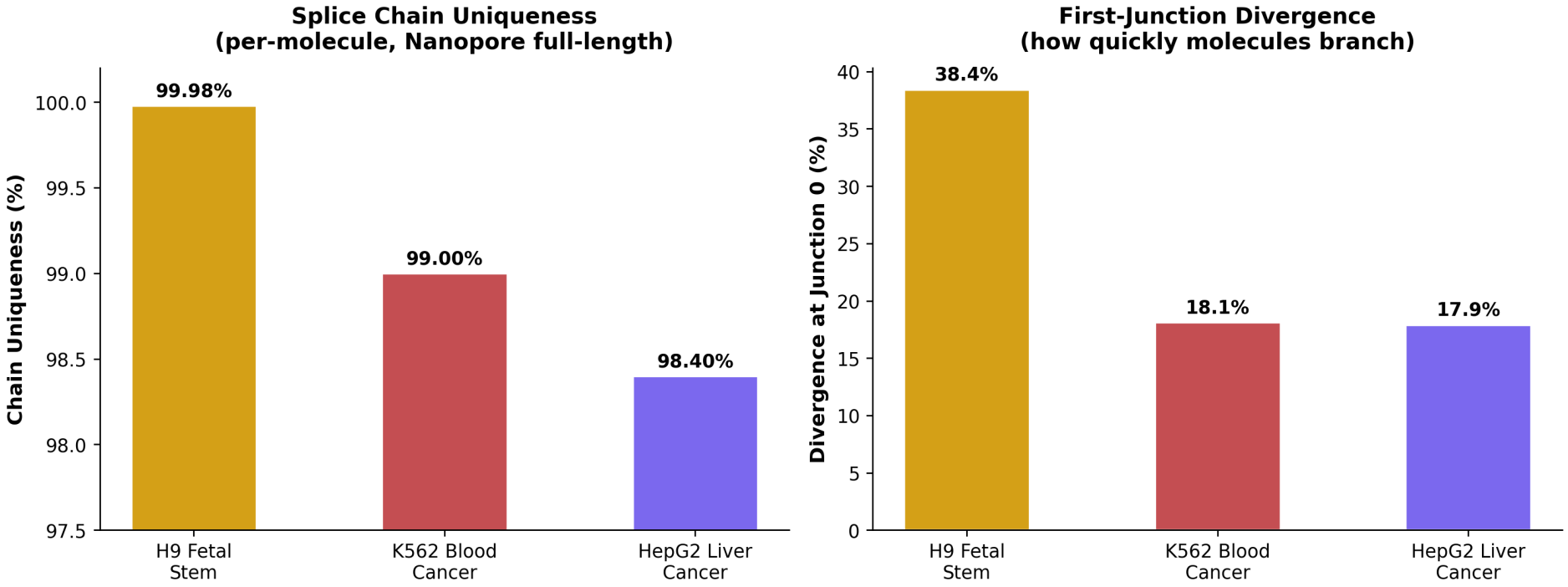
Figure 2. Splice Junction XOR Shift: Fetal vs Cancer



7M+ junctions from Nanopore full-length transcripts (SGNex, SRA public). Fetal (gold) favors transitions. Cancer (red/purple) shifts +7.4% toward identity.

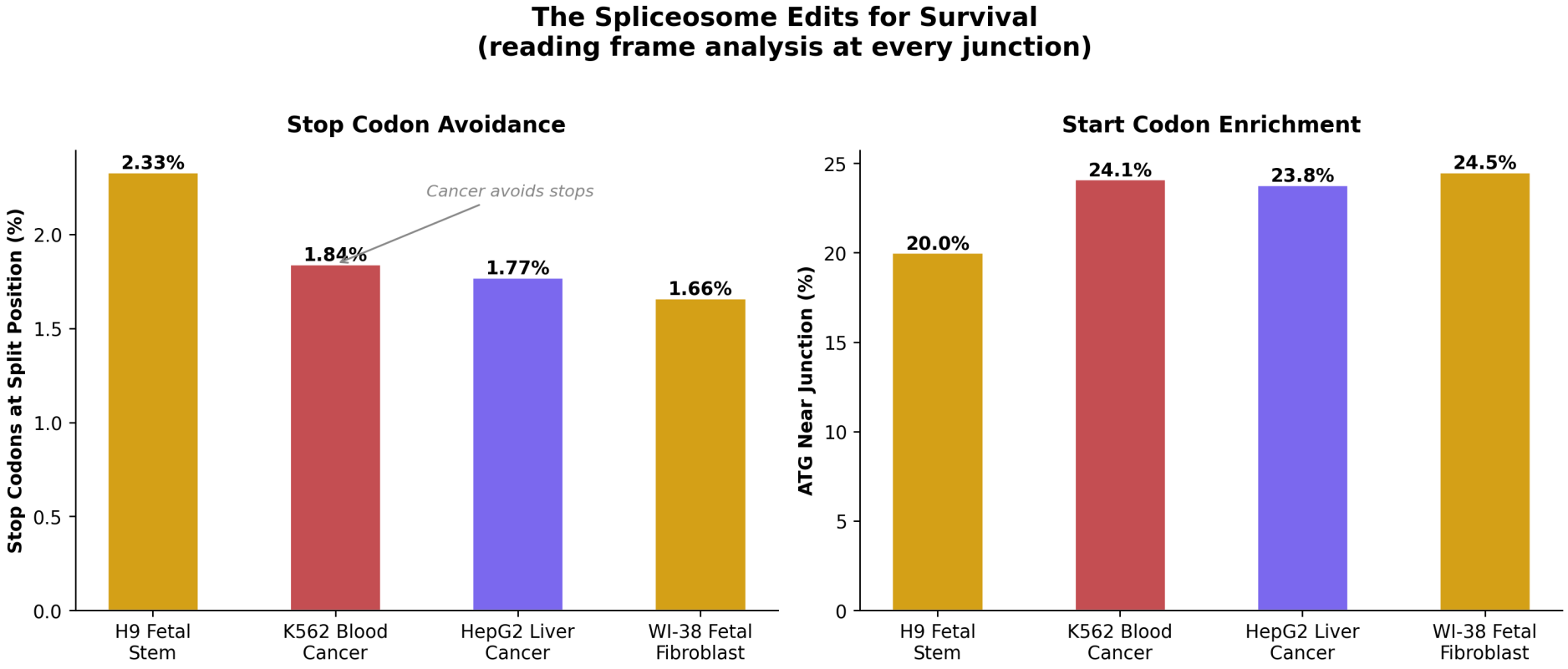
Figure 3. Splice Chain Uniqueness and First-Junction Divergence

Fetal Cells Explore. Cancer Converges.



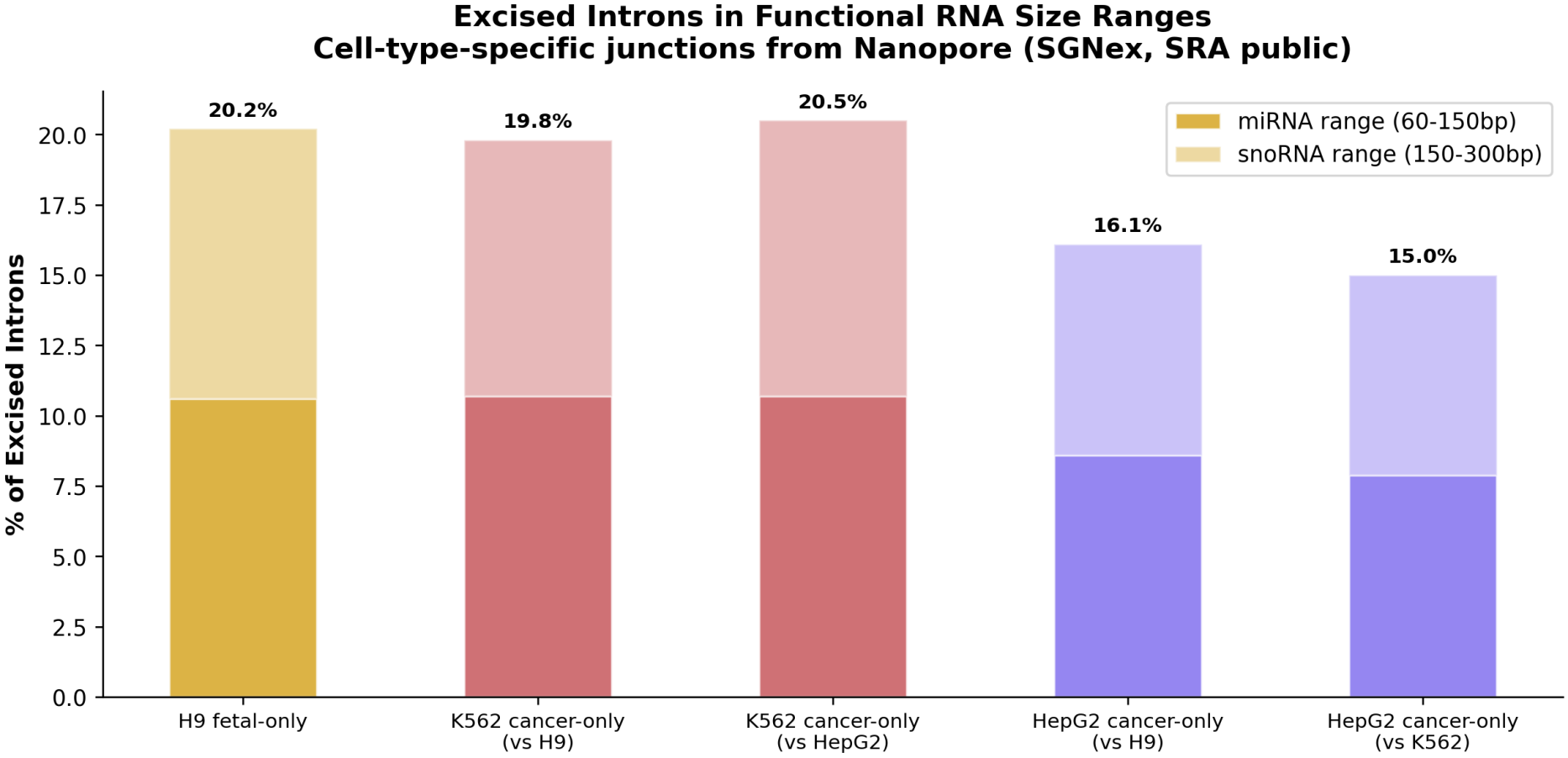
Per-molecule analysis. Fetal: 99.98% unique chains, 38.4% divergence at junction 0. Cancer: 98.4% unique, 18.1% divergence. Fetal explores. Cancer converges.

Figure 4. Stop Codon Avoidance and Start Codon Enrichment



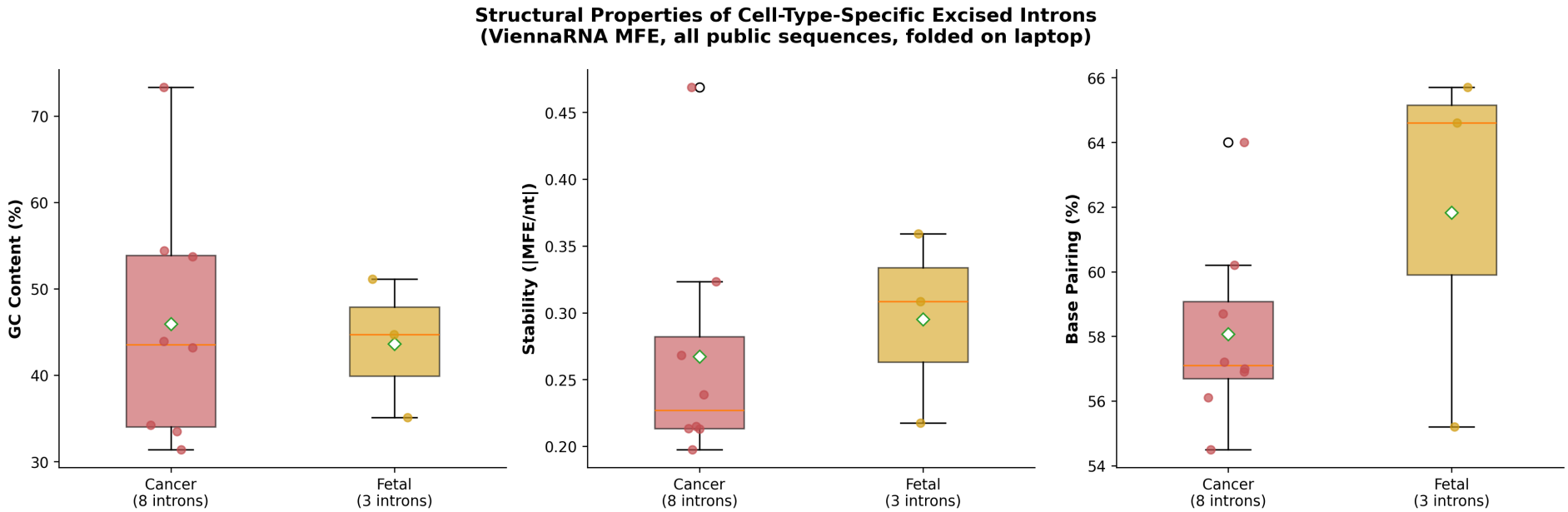
Reading frame analysis at every splice junction. Cancer avoids stops (1.66-1.84% vs 2.33% fetal) and enriches for ATG start codons (24.5% vs 20.0%). The spliceosome edits for survival.

Figure 5. Excised Introns in Functional RNA Size Ranges



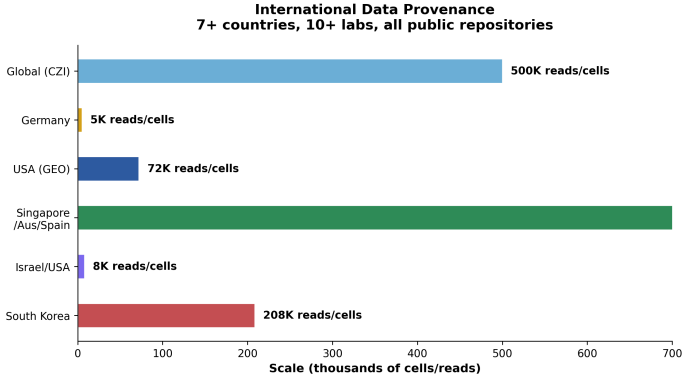
10-20% of cell-type-specific excised introns are in miRNA (60-150bp) or snoRNA (60-300bp) range. Cell-type-specific junctions from Nanopore (SGNex, SRA public).

Figure 6. Structural Properties of Cancer vs Fetal Excised Introns



11 introns folded with ViennaRNA 2.7.2 on personal laptop. Cancer (red, 8 introns) vs fetal (gold, 3 introns). HBA intron: 73.3% GC, -0.469 kcal/mol/nt. ALB intron: 33.5% GC, -0.213 kcal/mol/nt.

Figure 7. International Data Provenance



2.0M reads/cells

7+ countries, 10+ labs, all public repositories. Korean NSCLC (208K cells), Israeli GBM (7.9K), Singapore Nanopore (2M+ reads), US GEO series, CellxGene Census (500K+).